Electricity In The Home

Read the following passage and answer the questions which follow.

Questions

1 Why do the wires in an electrical cord need to be thicker when connected to an electric room heater than when connected to a table lamp ?

2 How does the plug fuse reduce the risk of fire ?

3 What is the most common cause of a fuse “blowing’?

4 Why is the **lowest** value fuse wire chosen to protect a circuit carrying a normal current ?

5 Some electrical cords on appliances have two wires instead of three. Which wire would be missing on the “two wire” cord ?

6 In what circumstances is it safe to use a “two wire” cord instead of a “three wire” cord ?

7 What is the function of the earth wire ?

8 Why is the external casing of an electrical cord (flex) made of PVC, rubber or textile ?

9 A three wire cord has three wires capable of carrying an electric current. Name the three wires and describe the function of each wire.

10 From the table show, what plug fuse would you fit to appliances of power rating:

(a) 400 W (b) 2 kW (c) 100 W ?

11 From the table show, what thickness of flex would you connect to:

(a) an iron (b) a vacuum cleaner (c) a toaster

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Answers

1 An electric room heater draws more current than a table lamp so the wires may get hot if the current encounters resistance . Thicker wires offer less resistance and allow a greater current to flow without the hazard of the wires becoming hot and causing a fire.

2 The fuse in the plug melts and breaks the circuit if a fault develops.

3 A short circuit.

4 3 A

5 The earth wire (green and yellow)

6 When the casing of the appliance is made of an insulator.

7 If a fault occurs and the casing of an appliance becomes “live”, the earth wire will conduct the electricity to earth and prevent a person who touches the casing from receiving an electric shock.

8 These substances are insulators and so protect persons who handle the cord when electricity is flowing through the wires.

9 Earth wire - caries current to earth in the event of a fault caused by a short circuit for instance.
Neutral wire - carries to current from the appliance to the source of current
Live wire - carries the current from the source to the appliance. The switch which operates the appliance will always be installed on this section of the circuit.

10 (a) 3 A (b) 13 A (c) 3 A

11 (a) 0.75 mm^2 (b) 0.75 mm^2 (c) 0.75 mm^2